Selective Adversarial Causal Oversight: An Information-Partitioned Benchmark Contract and Countermodel-Grounded Reference Verifier

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Abstract

LLMs are increasingly used to evaluate causal claims, but real causal oversight is both selective and adversarial: claimants may hold private information, exploit non-identifiability, and present persuasive but incomplete arguments. We formalize this setting as selective adversarial causal oversight, where a verifier with only public evidence must decide whether a natural-language claim is valid, invalid, or unidentifiable. We present a benchmark contract and prototype synthetic core with schema-level public/gold partitioning, twelve graph families across Pearl's hierarchy, structurally grounded attacks, a persuasion overlay, and OOD buckets for graph, mechanism, attack-family, context-shift, and paired-flip shifts. We also propose countermodel-grounded selective verification, a reference pipeline that parses the claim, builds an assumption ledger, searches for observationally compatible countermodels, and returns an abstention-aware verdict. On a fixed-code exploratory snapshot (three seeds, ten samples per family, difficulty 0.55), the reference verifier improves the accuracy/unsafe-acceptance/wise-refusal tradeoff over public-view heuristic baselines, while a deliberately leaking positive control shows how oracle information can inflate benchmark scores. These results support the need for public/gold partitioning in causal oversight benchmarks; larger synthetic runs, real-grounded cases, full API-based model baselines, and human audit remain required before claiming a mature benchmark release.

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1. Introduction

LLMs are now used as judges, debate moderators, and policy explainers, but evaluating their causal reasoning remains awkward. Existing benchmarks largely treat causal questions as graded multiple-choice problems (Jin et al., 2023; 2024; Frohberg & Binder, 2022): the evaluator inspects an oracle structural causal model, fixes a yes/no answer, and scores the LLM's prediction against it. This format is convenient but masks two features that dominate causal oversight in practice.

Information asymmetry. A claimant – a clinician, policy advocate, research summarizer, or deceptive agent – often has access to private information that the verifier does not: the data-generating process, hidden confounders, selection mechanisms, or simply domain familiarity. The verifier sees only public evidence and the claim text. Ignoring this asymmetry collapses a hard oversight problem into an easy closed-world recall problem.

Wise refusal. A core lesson of causal inference is that, given a fixed public evidence base, many queries are not identifiable (Pearl, 2009; Bareinboim et al., 2022). In such cases the correct behaviour is not to commit to a verdict, but to abstain with a reason. Modern LLM judges, however, are trained to commit and exhibit sycophancy and position bias when uncertain (Sharma et al., 2024; Chen et al., 2024; Ye et al., 2024). Existing causal benchmarks rarely measure "did the model refuse for the right reason" as a first-class quantity.

Our task. We formalize selective adversarial causal oversight (Section 3). An attacker generates a natural-language claim with full knowledge of the gold structural causal model (SCM); a verifier sees only the public projection of the scenario plus the claim text; and the evaluator scores the verifier's three-way verdict $\{\text{valid}, \text{invalid}, \text{unidentifiable}\}$ together with a structured assumption ledger, refusal reason, and witness.

Our benchmark contract. We construct a prototype synthetic core of twelve graph families that explicitly mark family-level identifiability, observationally equivalent counter-structures, and a finite catalogue of structural attack modes. A persuasion overlay projects each claim into authority, expert-tone, confidence, consensus and concealment styles. Splits enforce partitioned holdouts at the family, mechanism, attack-family, context-shift, and paired-flip levels, and leakage-prone metadata fields are stripped from the verifier-side view by construction (Section 4). Following our current research plan, the literature-grounded subset and dual human audit are treated as required completion gates rather than finished evidence.

Our reference verifier. Countermodel-grounded selective verification (Section 5) is a four-stage pipeline that (i) parses the claim into structured slots, (ii) builds an assumption ledger of identifying preconditions, (iii) searches for observationally compatible counter-SCMs that disagree with the claim on the target query, and (iv) consults Pearl-style identification tools as supporting evidence. A fixed decision rule turns these artifacts into one of the three verdicts plus an abstention reason and a missing-information specification.

Our claims. Three propositions (Section 6) tie selective abstention, countermodel witnesses, and oracle leakage to formal properties of the task. Empirically (Section 7), on the current three-seed, ten-samples-per-family workspace snapshot we observe that (a) the full system improves verdict accuracy over Direct Judge from 0.504 to 1.000 on test_iid and from 0.161 to 0.891 on test_ood, while driving unsafe acceptance from 0.89/0.97 to 0.00/0.00; (b) a deliberate oracle-leaking positive control reaches 1.00 accuracy on both dedicated attack-only splits and therefore remains a useful positive control for any benchmark that does not enforce a public/gold partition; and (c) removing abstention preserves high IID raw accuracy (0.856) but collapses unidentifiable-awareness to 0.00 on both splits, which is exactly the failure mode the task is designed to surface. We discuss limitations – in particular, the still-modest benchmark scale, missing real-grounded human audit, and API model-baseline status – in Section 7 and Section 8.

2. Related Work

Causal benchmarks for LLMs. CLadder (Jin et al., 2023) introduces a 10K-sample symbolic causal benchmark spanning Pearl's three rungs, with ground truth derived from an oracle inference engine. Corr2Cause (Jin et al., 2024) probes whether LLMs can convert correlational statements into causal structure; the authors report near-random performance and a sharp drop under perturbation. CRASS (Frohberg & Binder, 2022) targets counterfactual conditionals, and CausalReasoningBenchmark (Sawarni et al., 2026) separates causal identification from numerical estimation on real datasets. We differ from these benchmarks in three structural ways: (i) we strip oracle fields from the verifier's view by schema construction rather than by prompt hygiene, (ii) we elevate unidentifiable to a first-class supervised label tied to public evidence, and (iii) we evaluate not just accuracy but a wise-refusal-aware metric bundle and a deliberately leaking positive control.

Fact verification and real-grounding sources. CHECKWHY (Si et al., 2024) studies causal fact verification with claim-evidence-argument triplets and support/refute/not-enough-info labels, while SciFact (Wadden et al., 2020) and HealthVer (Sarrouti et al., 2021) focus on scientific and health claim verification. Clinical evidence corpora such as Evidence Inference 2.0 (DeYoung et al., 2020) and EvidenceOutcomes (Zhou et al., 2025) provide treatment-comparator-outcome language, and causal/uplift datasets such as ACIC challenges (Atlantic Causal Inference Conference, 2019), RealCause (Neal et al., 2021), Criteo Uplift (Criteo AI Lab, 2014), KuaiRec/KuaiRand (Gao et al., 2022a;b), and the Open Bandit Dataset (Saito et al., 2021) provide grounded settings with interventions, exposure policies, or logged decisions. We use these as target sources for the planned real-grounded subset, but their native labels do not directly equal our $\{\text{valid}, \text{invalid}, \text{unidentifiable}\}$ causal identifiability contract and therefore require re-annotation.

Selective prediction and abstention. Classical selective prediction (Geifman & El-Yaniv, 2017; 2019) introduces a reject option to deep classifiers. Recent work re-derives this idea for LLMs and finds that abstention quality is sensitive to calibration (Guo et al., 2017; Wen et al., 2025). Our task definition is closest to this literature in spirit: the verifier explicitly outputs an abstention reason when public evidence does not determine the query, for example observational equivalence or missing identifying support.

LLM-as-judge and sycophancy. LLM judges show systematic biases: sycophancy (Sharma et al., 2024), position and verbosity bias (Ye et al., 2024), and authority effects (Chen et al., 2024). Causal claims compound these biases because the surface plausibility of an explanation is correlated with the structural correctness of the underlying identification. Our persuasion overlay is designed to expose exactly that gap.

Scalable oversight and debate. Irving et al. (2018) and Bowman et al. (2022) frame oversight as a game between informed agents and a less-informed judge. Our setting is a specialization of that frame to causal claims, with a structured rather than free-form transcript and an explicit abstention option for the judge.

Causal identification and tools. We build on canonical identification machinery: do-calculus and the backdoor/frontdoor criteria (Pearl, 2009; Shpitser & Pearl, 2008), LATE-style instrumental-variable identification (Imbens & Angrist, 1994), and probability-of-causation results (Tian & Pearl, 2000). Tool execution in our pipeline uses estimators in the spirit of DoWhy (Sharma & Kiciman, 2020).

3. Task: Selective Adversarial Causal Oversight

3.1. Information Partition

A scenario is generated as a structural causal model $\mathcal{M} = (\mathbf{V}, \mathbf{U}, F, P(\mathbf{U}))$ over observed variables $\mathbf{V}$ and hidden variables $\mathbf{U}$. We expose three contracted views:

- Attacker view: the gold SCM, the true DAG, hidden variable identities, full data, and a generator metadata bundle. The attacker uses this to craft a claim that is rhetorically plausible against public evidence but structurally constrained.
- Verifier view: a PublicCausalInstance containing only observed variables, public observed data, optional named role hints (proxy/instrument/mediator/selection), and the natural-language claim.
- Evaluator view: gold + verifier views, used solely to score the verdict and audit the witnesses.

The contract is enforced at the schema level: the PublicCausalInstance dataclass strips gold-only fields such as the true DAG, hidden variables, full SCM, full data, and gold labels, and sanitizes public text against explicit leakage patterns before the verifier-side view is emitted.

3.2. Output and Label Spaces

For a parsed claim with target (X, Y) over query type $q \in \{\text{association, intervention, counterfactual}\}$, the verifier outputs

- a verdict $\hat{y} \in \{\text{valid, invalid, unidentifiable}\}$,
- an identification status s that is either identified, contradicted, or underdetermined, consistent with $\hat{y}$,
- a refusal reason r when $\hat{y} = \text{unidentifiable}$,
- an assumption ledger of typed entries (supported, contradicted, unresolved),
- witness slots for supporting evidence, countermodels, and assumptions, and
- a missing-information spec listing the unsupported assumptions and the evidence that would close the identification gap.

3.3. Why the Third Label is Necessary

For three of our twelve graph families (l1_latent_confounding, l2_invalid_iv, l3_counterfactual_ambiguity), there exist multiple SCMs $\mathcal{M}_1, \mathcal{M}_2$ that are observationally equivalent but disagree on the target query. Pearl's causal hierarchy theorem (Bareinboim et al., 2022) formalizes the general case: layer-k data underdetermines layer-$(k+1)$ truth almost everywhere. We therefore treat unidentifiable not as a degraded label but as the unique correct output when public evidence does not pin down a single query value.

4. Benchmark

4.1. Synthetic Core: Twelve Graph Families

The benchmark generator instantiates each scenario from a registry of twelve deterministic graph family blueprints (Table 1), distributed across the three rungs of Pearl's hierarchy (four L1, five L2, three L3 families). Each family is tagged with a family-level identifiability status (identifiable vs. potentially_unidentifiable), a list of supported gold labels, and a catalogue of attack modes that match the family's structure.

4.2. Persuasion Overlay

A separate overlay layer projects each structurally-fixed claim into a persuasion-styled variant. The taxonomy mirrors the social-science persuasion literature (Zeng et al., 2024) but is deliberately narrowed to five axes: authority, expert_tone, confidence, consensus, and concealment. The paper-facing primary axis uses four of these and reports expert_tone only in the taxonomy-complete artifact, to avoid double-counting tone effects with authority effects in the main table.

Table 1. Twelve graph families across Pearl's hierarchy. Each family fixes observable roles (treatment, outcome, context, adjuster, instrument, mediator, proxy, selection, latent confounder), a supported label set, and an attack catalogue. "ID" = structurally identifiable; "PU" = potentially unidentifiable.

Rung	Family	ID/PU	Supported Labels	Primary Attack Modes
L1	latent_confounding	PU	invalid, unidentifiable	assoc. overclaim, hidden-conf. denial
L1	selection_bias	PU	invalid, unidentifiable	selection-bias obfuscation
L1	proxy_disambiguation	ID	valid, invalid	assoc. overclaim
L1	reverse_causality	PU	invalid, unidentifiable	assoc. overclaim
L2	valid_backdoor	ID	valid, invalid	invalid-adjustment, heterogeneity overgen.
L2	valid_iv	ID	valid, invalid	heterogeneity overgen.
L2	invalid_iv	PU	invalid, unidentifiable	weak-IV-as-valid-IV, exclusion claim
L2	subgroup_heterogeneity	PU	invalid, unidentifiable	heterogeneity overgen.
L2	frontdoor_partial_meas.	PU	valid, unidentifiable	unidentifiable disguised as valid
L3	counterfactual_ambiguity	PU	unidentifiable	counterfactual overclaim
L3	mediation_abduction	ID	valid, invalid	function-form manipulation
L3	monotonicity_cross_world	PU	invalid, unidentifiable	cross-world failure, function-form

4.3. Witnesses

Every gold sample carries up to three structured witnesses: a support witness (why the gold label holds under public evidence), a countermodel witness (an observationally compatible alternative SCM that disagrees with the claim on the target query), and an assumption witness (which identifying assumption is unsupported or contradicted). These witnesses are first-class evaluation objects rather than narrative decoration: they ground the decision rule in Section 5 and the qualitative witness-faithfulness probe in Section 7.

4.4. Splits and Holdouts

The split manifest enforces non-overlapping train, dev, test_iid, and test_ood subsets. The holdout strategy explicitly tracks several orthogonal axes: graph family, mechanism (e.g. confounding vs. collider), attack-family, context-shift domain, paired-flip pairs, plus lexical/template and variable-renaming controls inside the generator. This goes beyond the lexical-perturbation OOD studied in prior work (Jin et al., 2024), which we view as a useful but insufficient stress test. Section 7 reports the five paper-facing OOD buckets that correspond directly to the benchmark contract: graph-family, mechanism, attack-family, context-shift, and paired-flip OOD.

4.5. Real-Grounded Subset

A literature-grounded subset (target size 150–300 cases) is curated from economics, policy, epidemiology and education observational studies. Each case records a source citation, a public-evidence summary, the visible-vs-hidden information contract, the gold label, identifying assumptions, and a witness note. The synthetic and real-grounded subsets share the same claim schema, so models trained or tuned on the synthetic core can be zero-shot evaluated on the real-grounded subset. This subset is part of the benchmark design; the current workspace also includes a 150-case human-audit package generator, but not yet dual-annotator agreement results. Full real-grounded evaluation is therefore reserved for the human-audit phase (see Section 8).

5. Method: Countermodel-Grounded Selective Verification

The verifier pipeline runs four sequential stages and emits a single VerifierDecision object.

Stage 1: Claim parsing. Given a public scenario and the natural-language claim, the parser extracts treatment, outcome, query type, polarity, strength, mentioned and implied assumptions, the rhetorical strategy (e.g. an adjustment-sufficiency claim or an IV appeal), and an abstention-risk flag derived from claim-level cues.

Stage 2: Assumption ledger. The ledger initializes one entry per identifying precondition required by the parsed claim. Status starts as unresolved and is updated by later stages: tool-backed evidence may flip an entry to supported, contradictory evidence flips it to contradicted. Baseline assumptions (consistency, positivity) are tracked but not counted as identifying support.

Stage 3: Countermodel search. The countermodel module enumerates structurally-compatible alternative SCMs for the public scenario and scores each on

(a) the public-equivalence distance defined in Section 6, (b) disagreement with the claim on the target query, and (c) which identifying assumption it triggers. The module returns a chosen candidate, a short explanation, and a verdict suggestion.

Stage 4: Tool-backed adjudication. When no countermodel survives, a tool runner executes the appropriate identification tools for the rung – conditional independence and association checks for L1, backdoor, frontdoor and IV estimators for L2, and SCM-based counterfactual reasoning plus probability-of-necessity bounds for L3 (Tian & Pearl, 2000; Sharma & Kiciman, 2020). Tool outputs are stored as a structured trace that records which ledger entries are supported or contradicted.

Decision rule. The four stages are combined by a fixed rule:

1. If a strong countermodel survives and the claim's verdict suggestion is invalid, return invalid.
2. Else if a countermodel survives and the candidates disagree on the target query, return unidentifiable with refusal reason observational_equivalence.
3. Else if any identifying assumption is contradicted by tools, return invalid.
4. Else if any core identifying assumption remains unsupported or no primary-supporting tool record exists, return unidentifiable with a missing-identifying-support reason; when the abstention gate fires on a high-risk claim, we return the stricter primary-support variant.
5. Else return valid.

This rule is deterministic, auditable, and is implemented in fewer than 300 lines of Python; the closed-form schema makes selective verdicts directly comparable across systems.

6. Theoretical Guarantees

We make three minimal claims that pin down why the task design is necessary rather than aesthetic.

Proposition 6.1 (Abstention Necessity). Let E denote the public evidence and q the target query. Suppose there exist SCMs $\mathcal{M}_1, \mathcal{M}_2$ that both induce E as their observational marginal but disagree on q. Then any verifier that does not admit unidentifiable as an output produces an incorrect verdict on at least one of $\mathcal{M}_1, \mathcal{M}_2$.

Proof sketch. Both SCMs are public-evidence-equivalent, so the verifier's input distribution is identical under either generative model. Any committed verdict is therefore the same under both, but $q(\mathcal{M}_1) \neq q(\mathcal{M}_2)$, so the verdict is wrong under exactly one. □

Empirical correspondence. Section 7 (No-Abstention ablation) instantiates this proposition: removing the abstention head retains 0.856/0.462 raw accuracy on IID/OOD but zeroes out unidentifiable-awareness.

Definition 6.2 (ε-public equivalence). Fix a public projection $\Pi(\mathcal{M})$ that contains exactly the fields available to the verifier, and a finite public evidence map $\phi(\Pi(\mathcal{M}))$ containing the symbolic public fields plus the observable summary statistics used by the verifier. Two SCMs are ε-public-equivalent, written $\mathcal{M}_1 \equiv_{\Pi,\varepsilon} \mathcal{M}_2$, when their symbolic public fields match and $\|\phi(\Pi(\mathcal{M}_1)) - \phi(\Pi(\mathcal{M}_2))\|_\infty \leq \varepsilon$ on the numeric public summaries.

Proposition 6.3 (Countermodel Witness Soundness). Suppose a verifier exhibits an SCM $\mathcal{M}'$ such that $\mathcal{M}' \equiv_{\Pi,\varepsilon} \mathcal{M}$ for the benchmark's declared public evidence map, but $q(\mathcal{M}') \neq q(\mathcal{M})$ for the target query. Then no verifier whose input is restricted to the ε-public-equivalence class can uniquely identify the value of q. A countermodel witness is therefore a certificate of non-identifiability relative to (Π, ϕ, ε), not a claim about all possible evidence outside the public contract.

Proof sketch. The verifier's admissible input is invariant over the ε-public-equivalence class, while the query value differs across two members of that class. Any rule that commits to a unique value of q from that input alone must be wrong for at least one member. The qualitative shuffle/delete probe in Section 7 provides exploratory empirical support that these witnesses are load-bearing in the implemented pipeline. □

Proposition 6.4 (Oracle Leakage Inflation). Let V_{public} be any fixed verifier over public inputs, and let $g \in$ GOLD_ONLY_FIELDS contain a decodable label signal. Define an oracle-positive-control verifier $V^\star_{\text{leak}}$ that uses V_{public} when the decoded signal is unavailable and otherwise returns the decoded label. Then $\mathbb{E}[\mathrm{Acc}(V^\star_{\text{leak}})] \geq \mathbb{E}[\mathrm{Acc}(V_{\text{public}})]$, with strict inequality whenever the decoded signal corrects at least one expected error of V_{public}. This is not a monotonicity statement about arbitrary verifiers with extra fields; it is a positive-control construction.

Empirical correspondence. Section 7 (Leakage Study) instantiates this proposition with a positive-control verifier; the artifact's supports_warning field operationalizes the proposition as a benchmark-level contamination gate.

The role of Theorem 6.4 is operational: it justifies our leakage study (Section 7) by predicting that any inflation observed there is, by construction, leakage rather than oversight ability.

7. Experiments

7.1. Experimental Setup

We report the current workspace snapshot: three seeds with ten samples per family across all twelve graph families. The main benchmark runs at difficulty 0.55 and compares the full countermodel-grounded verifier against public-view heuristic judge/debate, tool-only and refusal-aware baselines. The component ablation suite replaces one or more pipeline stages with a no-op surrogate. We also include a dedicated attack-only leakage study under the hidden-information-aware profile. These runs are reproducible from the current repository state. All paper-facing tables in this draft are generated by python -m experiments.paper_artifacts from the hashed bundle recorded in outputs/paper_facing/manifest.json. They are not a saturation study: the synthetic core is still modest, the leakage study is exploratory because it uses ten rather than the pre-registered sixty samples per family, and the real-grounded and human-audit phases are not yet complete. We also include two live DashScope qwen2.5-7b-instruct API plumbing smoke runs covering the available IID smoke slice ($n = 9$) and an OOD smoke slice ($n = 12$); they record prompts, raw responses, model settings, and fallback status, and are not treated as a full API baseline matrix. We therefore present the empirical results as fixed-code evidence for the benchmark contract and reference verifier, not as a final benchmark release.

7.2. Main Benchmark

Table 2 compares the full Countermodel-Grounded verifier against compact public-view heuristic baselines on test_iid and test_ood. These baselines are deterministic runner implementations named after common judge/debate prompting patterns; they are not claimed to be strong API LLM baselines. We report the three metrics most central to our task definition: verdict accuracy, unsafe-acceptance rate, and wise-refusal recall. Self-Consistency Judge matches CoT Judge exactly in this snapshot and is omitted from the compact table for readability; the full artifact retains it.

Three observations matter. First, Direct Judge exhibits exactly the failure mode that motivates the task: extremely high unsafe acceptance (0.89 IID, 0.97 OOD) under public evidence constraints. Second, CoT Judge, Debate Baseline, and the explicit Refusal-Aware baseline drive unsafe acceptance to zero, but do so by over-refusing: on test_ood, their over-refusal rates are 0.850, 0.677, and 0.557, respectively, versus 0.010 for the full system. Third, the near-perfect IID score should be read as a reference-verifier sanity check on the synthetic contract, not as evidence that generic LLMs have solved causal oversight. The stronger claim is the tradeoff shape: the full pipeline maintains zero unsafe acceptance while preserving OOD wise-refusal recall and substantially improving raw verdict accuracy over public-view heuristic baselines.

7.3. Component Ablation: What Drives Wise Refusal?

Table 3 extends the ablation to four pipeline components, including a No-Abstention variant that disables the third-label output entirely. Two patterns are especially important. First, No-Abstention preserves high IID raw accuracy (0.856) but zeroes out unidentifiable-awareness on both splits, which is exactly the failure mode the selective task is designed to surface. Second, removing the countermodel stage hurts both OOD verdict accuracy and OOD unidentifiable-awareness (0.504 and 0.591). The current snapshot is not a strict monotone ranking across every metric: No-Tools exceeds the full system on OOD raw accuracy (0.911 vs. 0.891) and unidentifiable-awareness (0.964 vs. 0.825), but does so with substantially higher over-refusal (0.150 vs. 0.010) and lower wise-refusal precision (0.877 vs. 0.989). We therefore interpret the full system as the more balanced reference verifier, not as uniformly best on every scalar metric.

7.4. Leakage Study

The current leakage study is explicitly exploratory because it uses only ten samples per family. We therefore present it as a positive-control sanity check that the public/gold contract behaves as Proposition 6.4 predicts, rather than as a definitive estimate of contamination magnitude. Full leakage evidence requires increasing samples_per_family to at least 60.

We instantiate two verifiers on a dedicated attack-only benchmark with the hidden-information-aware attack profile: a clean public verifier and a deliberately oracle-leaking control that decodes benchmark-answer supervision from a protocol-violating public metadata partition. Per Theorem 6.4, any positive gap is by construc-

Table 2. Main benchmark, current mainline snapshot (three seeds, ten samples per family, difficulty 0.55). Mean ± std across seeds; full 95% CIs, calibration metrics, and over-refusal rates are reported in outputs/review_fixed_exp_main_benchmark.md.

System	Split	Verdict Acc.	Unsafe Accept	Wise Refusal Recall
Direct Judge	test_iid	0.504 ± 0.090	0.889 ± 0.192	0.500 ± 0.500
Direct Judge	test_ood	0.161 ± 0.056	0.970 ± 0.029	0.184 ± 0.033
CoT Judge	test_iid	0.322 ± 0.019	0.000 ± 0.000	1.000 ± 0.000
CoT Judge	test_ood	0.592 ± 0.053	0.000 ± 0.000	1.000 ± 0.000
Tool Baseline	test_iid	0.537 ± 0.116	0.372 ± 0.256	0.667 ± 0.577
Tool Baseline	test_ood	0.346 ± 0.092	0.423 ± 0.057	0.559 ± 0.067
Debate Baseline	test_iid	0.322 ± 0.019	0.000 ± 0.000	1.000 ± 0.000
Debate Baseline	test_ood	0.456 ± 0.040	0.000 ± 0.000	0.610 ± 0.025
Refusal-Aware	test_iid	0.752 ± 0.045	0.000 ± 0.000	1.000 ± 0.000
Refusal-Aware	test_ood	0.504 ± 0.090	0.000 ± 0.000	0.591 ± 0.020
Countermodel-Grounded	test_iid	1.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000
Countermodel-Grounded	test_ood	0.891 ± 0.005	0.000 ± 0.000	0.825 ± 0.018

Table 3. Component ablation. Unidentifiable-awareness is the task-critical third-label measure; the full artifact includes precision, over-refusal, calibration, and countermodel coverage.

System	Split	Acc.	Unident. Awareness
Full	iid	1.000	1.000
Full	ood	0.891	0.825
No-Ledger	iid	0.144	1.000
No-Ledger	ood	0.520	1.000
No-Countermodel	iid	0.752	1.000
No-Countermodel	ood	0.504	0.591
No-Abstention	iid	0.856	0.000
No-Abstention	ood	0.462	0.000
No-Tools	iid	0.570	1.000
No-Tools	ood	0.911	0.964

Table 4. Leakage study. Clean public verifier vs. an oracle-positive control with protocol-violating public metadata. Any positive gap is leakage inflation (Theorem 6.4). Numbers from the accompanying outputs/review_fixed_exp_leakage_study.md artifact.

Split	Clean Acc.	Leaking Acc.	Δ	Clean 95% CI	Leaking 95% CI
test_iid	0.579	1.000	+0.421	$[0.167, 1.000]$	$[1.000, 1.000]$
test_ood	0.845	1.000	+0.155	$[0.779, 0.912]$	$[1.000, 1.000]$

tion leakage inflation. Table 4 shows a gap of +0.421 on test_iid and +0.155 on test_ood. Both paired tests support the warning in this small run (bootstrap $p = 0.0395/0.0015$, McNemar $p = 0.0156/0.0000$ before the table's rounding), but the protocol notice still prevents treating this as formal leakage evidence until the sample count is increased. ECE drops from 0.400 to 0.000 on test_iid and from 0.077 to 0.000 on test_ood, illustrating how leaked supervision can masquerade as calibration.

Table 5. Adversarial robustness. Mean ± std across seeds; full results are reported in the accompanying exp_adversarial_robustness.md artifact.

Strength	Split	Acc.	Wise Refusal Recall
Weak	iid	0.889 ± 0.192	0.667 ± 0.577
Weak	ood	0.845 ± 0.039	0.685 ± 0.082
Medium	iid	0.889 ± 0.192	0.667 ± 0.577
Medium	ood	0.845 ± 0.039	0.685 ± 0.082
Strong	iid	0.889 ± 0.192	0.667 ± 0.577
Strong	ood	0.845 ± 0.039	0.685 ± 0.082
Leak-Aware	iid	0.579 ± 0.417	0.667 ± 0.577
Leak-Aware	ood	0.845 ± 0.067	0.844 ± 0.104

7.5. Adversarial Robustness

Table 5 reports robustness across four attack-strength regimes: weak, medium, strong, and a leak-aware hidden-information profile. On test_ood, the verifier stays in a narrow 0.845 band across all four regimes while maintaining zero invalid acceptance. The test_iid picture is less stable: the hidden-information-aware regime drops to 0.579 ± 0.417, making it the hardest current IID stress case. Because the weak/medium/strong rows are nearly identical in this snapshot, we treat this section as exploratory stress testing rather than a headline robustness claim.

7.6. OOD Buckets

The current OOD artifact evaluates the five buckets defined by the benchmark contract: graph-family, mechanism, attack-family, context-shift, and paired-flip OOD. The full system remains perfect on graph-family and mechanism OOD (1.000 accuracy), stays strong on attack-family (0.820) and context-shift (0.883), and drops sharply on paired-flip (0.439). The paired-flip bucket also has the largest macro-F1 gap (0.423), mak-

Table 6. OOD bucket breakdown for the full verifier. The rows are generated from outputs/paper_facing/ood_table_rows.tex; CIs are bootstrap 95% intervals over seed means.

Bucket	Acc.	Macro-F1	Unsafe Accept	Wise Refusal Recall	Acc. 95% CI
Graph-Family	1.000	0.667	0.000	1.000	[1.000, 1.000]
Mechanism	1.000	0.667	0.000	1.000	[1.000, 1.000]
Attack-Family	0.820	0.541	0.000	0.737	[0.800, 0.833]
Context-Shift	0.883	0.792	0.000	0.828	[0.862, 0.923]
Paired-Flip	0.439	0.485	0.000	0.714	[0.409, 0.455]

ing it the clearest current stress case for structural disambiguation. The current significance artifact reports uninformative $p = 1$ rows for these bucket gaps, so we avoid strong significance claims until the OOD estimand and sample count are upgraded.

7.7. Witness Faithfulness (qualitative)

We run a small witness-faithfulness probe by shuffling or deleting the countermodel witness in the verifier output and re-scoring. When the countermodel witness is shuffled across instances, the system's unidentifiable-awareness collapses on the families that rely on observational-equivalence reasoning (l1_latent_confounding_family, l3_counterfactual_ambiguity_family). This indicates that the witness is load-bearing rather than ornamental. A full quantitative study (witness shuffle/delete with paired-bootstrap 95% CIs across all twelve families) is reserved for future work.

8. Discussion and Limitations

What we are not claiming. We are not claiming a state-of-the-art result on a saturated benchmark, nor that LLM judges are uniformly bad at causal reasoning. The claim is structural: once the task enforces a public/gold contract and rewards correct abstention, the ranking of systems changes, and a verifier that takes this shape seriously offers a better accuracy/refusal tradeoff than verifiers that flatten the problem into a closed-world judgment task.

Limitations. The synthetic core is, by construction, a model of the world rather than the world itself. The real-grounded subset is small, and full human dual-annotation is in progress. The current mainline snapshot uses ten samples per family, so bucket-level estimates can still have wide CIs; the dedicated leakage study is explicitly exploratory; and the hidden-information-aware IID robustness slice is not yet stable enough to support strong quantitative claims. The API-backed model evidence is currently a one-sample smoke test for the runner and audit schema, not a competitive baseline. Larger runs, completed human audit, and a pre-registered API model matrix will be needed before any broad statement about absolute performance is defensible. The existing transfer matrix is likewise a pilot: it shows that family-swapped attacker/verifier profiles are testable inside the benchmark contract, but it is still too small to support model-family conclusions. We document these caveats in the artifact and in this paper.

Future work. Three priorities: (i) closing the real-grounded subset with dual annotation and Cohen's κ on the four primary audit fields; (ii) scaling the current transfer pilot into a full-budget cross-model study across verifier families and sizes; and (iii) replacing the API smoke with a pre-registered closed/open model baseline matrix using fixed prompts, raw-response logging, and fallback rejection.

9. Conclusion

Selective adversarial causal oversight is the right unit of analysis for LLM-based causal evaluation. By modelling information asymmetry, treating unidentifiable as a first-class label, partitioning public from gold at the schema level, and grounding the verifier in countermodel witnesses, we obtain a benchmark and a method whose comparison shape is meaningful even under current mainline budgets. The next step is scale: more samples, more graph families, real-grounded human-audited cases, and cross-model transfer.

Software and Data

The repository contains the benchmark generator, verifier pipeline, experiment runners, fixed-code result artifacts, and regression tests used for the current snapshot. The current paper-facing reproducibility package is outputs/paper_facing/: its manifest records source artifact hashes, configuration, seeds, generated table rows, poster metrics, and the available API smoke artifact. Each experiment emits a configuration JSON, a seed list, raw per-sample predictions, an aggregated metrics file, 95% CIs, paired-significance tests, and a markdown summary. The leakage study additionally emits a leakage-warning field that downstream consumers can use as a contamination check. The human-audit and full API-model baseline gates are explicitly pending rather than backfilled.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here. We note in particular that our adversarial framing is intended for evaluation and oversight rather than deployment: the persuasion overlay is a stress-test artifact for verifiers, not a recipe for deceptive content generation. The leakage decoder is isolated as a positive-control diagnostic so that contamination risks are measurable rather than hidden.

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A. Schema Excerpts

The following is an abridged view of the leakage-control contract. The full benchmark/schema.py file contains additional sanitizers and serialization helpers; only the contract-level constants are reproduced here.

```
GOLD_ONLY_FIELDS = (
    "true_dag", "hidden_variables", "true_scm",
    "full_data", "ground_truth", "gold_label",
)

VERIFIER_VISIBLE_FIELDS = (
    "scenario_id", "description", "variables", "proxy_variables",
    "selection_mechanism", "observed_data", "data",
    "causal_level", "sanitized_public_metadata",
)

class VerdictLabel(str, Enum):
    VALID = "valid"
    INVALID = "invalid"
    UNIDENTIFIABLE = "unidentifiable"

class IdentificationStatus(str, Enum):
    IDENTIFIED = "identified"
    CONTRADICTED = "contradicted"
    UNDERDETERMINED = "underdetermined"
```

B. Decision Rule (Pseudocode)

```
def decide_verdict(parsed, ledger, cm, tool_trace):
    # Stage 1: strong invalid countermodel
    if cm.found and cm.suggests_invalid:
        return INVALID

    # Stage 2: observational equivalence + query disagreement
    if cm.found and cm.query_disagreement:
        return UNIDENTIFIABLE(reason="observational_equivalence")

    # Stage 3a: contradicted assumption
    if any(e.contradicted for e in ledger):
        return INVALID

    # Stage 3b: missing or unsupported identification
    if any(e.unsupported for e in ledger) \
       or not tools_support_claim(tool_trace) \
       or (needs_explicit_support and no_supported_id):
        return UNIDENTIFIABLE(reason="missing_identifying_support")

    # Stage 3c: abstention gate on high-risk valid claims
    if abstention_gate_reasons(parsed, ledger, tool_trace):
        return UNIDENTIFIABLE(reason="missing_primary_identifying_support")

    # Stage 4: support
    return VALID
```

C. Persuasion Overlay Examples

A claim with structural label unidentifiable can be projected into five persuasion styles. The taxonomy-complete artifact includes expert_tone; the paper-facing primary axis collapses to the four remaining styles to avoid double-counting tone effects.

- Authority: “As established by the foundational paper of the field, the effect of X on Y is positive.”
- Confidence: “It is unambiguous, on any reading of the data, that X causes Y.”
- Consensus: “Every recent meta-analysis agrees that X raises Y.”
- Concealment: omits any mention of the latent confounder or selection mechanism that the public evidence cannot resolve.
- Expert tone: heavy technical jargon without a corresponding identification argument.

D. Reproducibility Notes

Each experiment runner emits a fixed bundle of artifacts:

```
<run>_config.json
<run>_seed_list.json
<run>_raw_predictions.jsonl
<run>_aggregated_metrics.json
<run>_ci.json
<run>.md    (paper-facing markdown summary)
```

The aggregator reports seed-mean estimates and paired bootstrap 95% CIs; the significance file applies Holm-Bonferroni correction within and across splits. The leakage runner additionally writes a supports_warning field that downstream consumers can use as a hard gate on contamination, in line with Theorem 6.4. The command python -m experiments.paper_artifacts freezes the paper-facing manifest and LaTeX rows in outputs/paper_facing/. The command python -m experiments.exp_api_baseline_smoke.run performs the API-smoke path and rejects mock fallback by default.